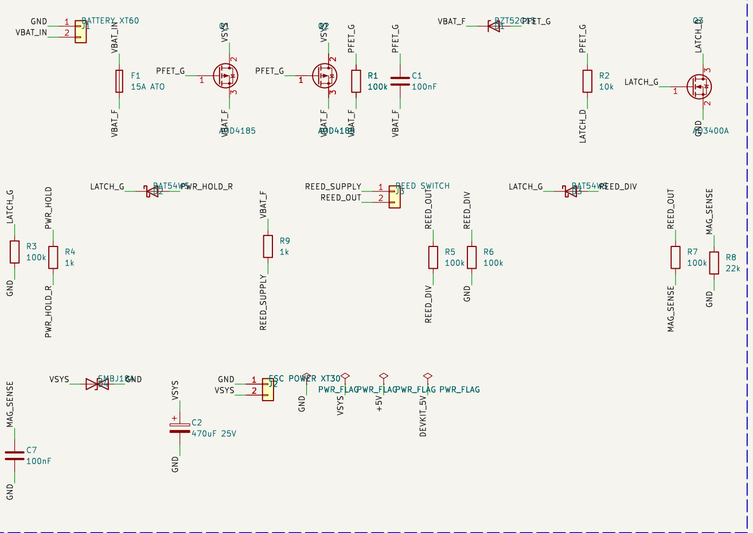
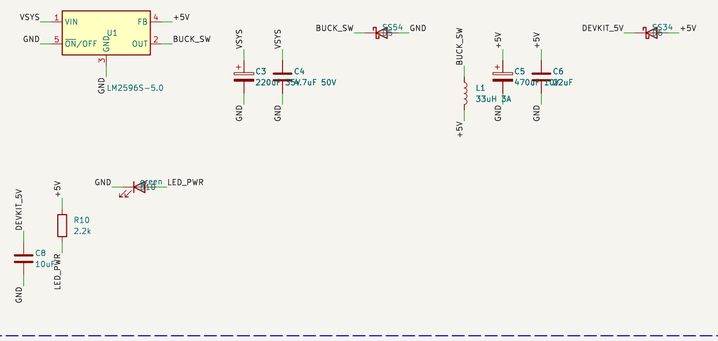
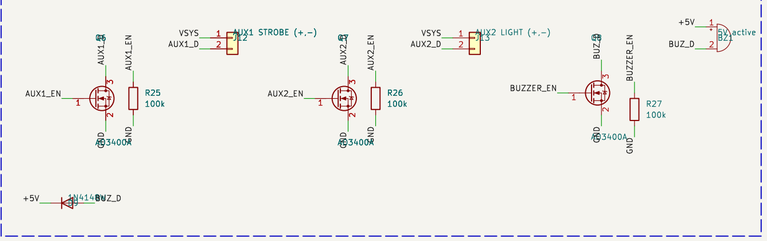
Battery input, 15 A fuse, magnet soft-latch power switch



5 V buck regulator (LM2596S-5)

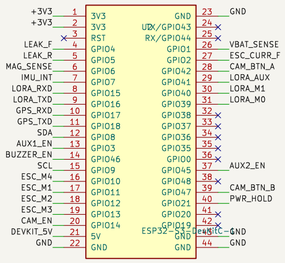


Switched outputs: strobe, spare, buzzer

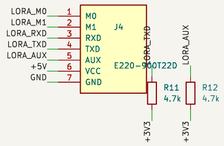


Generated from design.py by generate_board.py — edit design.py, not this file.
Magnet power switch: holding a magnet on the hull closes the reed switch -> Q3 pulls the P-FET gates low ->
the ESP32 boots and drives PWR_HOLD (GPIO21) high to keep power on. Shutdown = firmware releases PWR_HOLD.
All ADC inputs are on ADC1 (GPIO1-10) so they keep working while Wi-Fi talks to the camera.

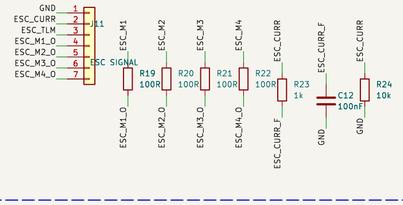
ESP32-S3-DevKitC-1 (plugs into two 22-pin sockets)



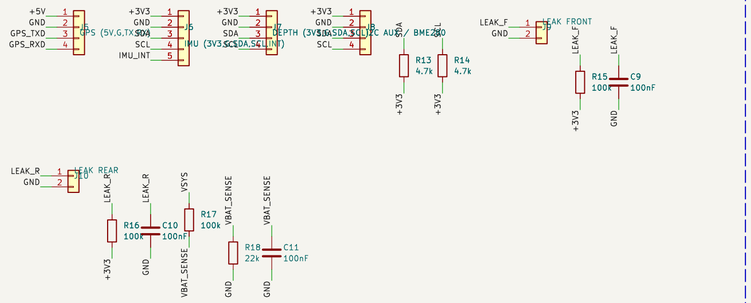
LoRa telemetry radio (Ebyte E220-900T22D, 868 MHz)



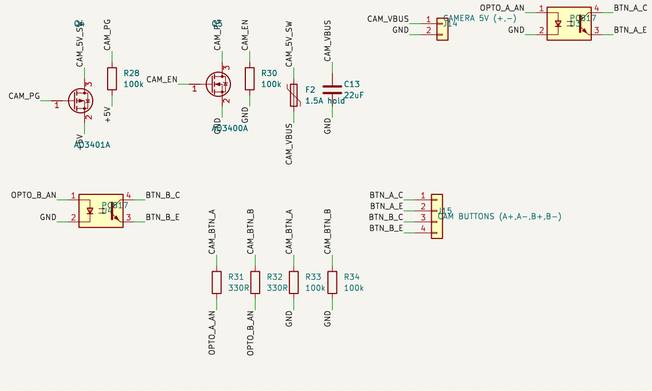
4-in-1 ESC signals (DShot) and current sense



Sensors: GPS, IMU, depth, leak, battery voltage



4K camera: switched 5 V and button opto-isolators



Test points and mounting holes



Single sheet. Nets are connected by name (labels).
ESP32-S3 DevKitC-1 carrier: 3S power latch, 5V buck, LoRa, 4-in-1 ESC, 4K camera control

BIOHUB / submarine

Sheet: /
File: auv-mainboard.kicad_sch

Title: Minnow AUV main board

Size: A2	Date: 2026-11-08	**Rev: 1.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1